A Lunar Terrain Facility for Robust and Resilient Learning-Based Autonomy

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***Abstract*—**In this paper, we present the Lunar Terrain Facility at the AstroX, Georgia Tech, a testbed for developing and validating autonomy algorithms and operation strategies for wheeled, legged, and hopping robots, with an emphasis on machine-learning (ML) and artificial-intelligence (AI) based autonomy that is multi-modal and adaptive to terrain and robust to illumination conditions. The facility includes a $5.8\ \mathrm{m} \times 4.0\ \mathrm{m} \times 0.30\ \mathrm{m}$ bed of 3–5 mm Merriam Crater basalt gravel with reconfigurable rock and relief features, four lamps with Fresnel optics that provide adjustable directional illumination, and a 16-camera motion-capture system that supplies a pose reference at 100 Hz. Reconfigurable Slip, sinkage, and shadow geometry support controlled experiments on granular terrain under directional illumination We describe the facility, the NOVA rover, the other platforms exercised on the same terrain, and the shared ROS 2 software architecture. We present a kinematic and actuation model of NOVA with a residual-compensated tracking control law and analyze its stability. We then present the following experiments conducted at the facility for various algorithm development: nonlinear trajectory tracking of NOVA under motion-capture feedback; on the Leo Rover, proprioceptive residual learning that reduced the residual-model error by 49–69% on held-out terrain, risk-aware navigation that completed five of sixteen trials and stopped safely in the other eleven, with no collisions, under injected sensing bias, and invariant adaptive fault estimation that detected 100% and identified 94% of fifty injected faults; humanoid and wheel-leg mobility on the gravel and on 10–30° slopes; and visual-inertial estimation failures under single-lamp low-angle illumination.

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1. INTRODUCTION

Future lunar surface systems are expected to perform increasingly complex tasks with limited supervision, including scientific surveying, infrastructure support, resource prospecting, logistics, and long-range exploration [1]. These tasks place coupled demands on perception, localization, mobility, planning, and control. The same patch of loose regolith changes the wheel–soil or foot–soil interaction, the appearance of the scene to a camera, the quality of the state estimate, and the traversability that a planner must reason about. Low solar elevation [2] adds long shadows, high local contrast, and abrupt changes in visual appearance that a machine-learned perception model trained on terrestrial data will not have seen [3], [4]. Autonomous operations at this level increasingly rely on ML and AI components, and the behavior of those components under lunar conditions cannot be established from terrestrial data alone. A ground-based facility that presents these granular and illumination conditions to a robot in a controlled and repeatable manner, and that measures the pose of the robot independently of the autonomy under test, provides a platform to develop and validate such autonomy algorithms, in the same way that a robotic spacecraft dynamics simulator with motion-capture ground truth [5] provides a platform for formation guidance, navigation, and control research.

Main motivation for the development of this facility is to test the autonomy that can be used on lunar surface missions in which robots of more than one locomotion mode operate on loose regolith under low-angle illumination. For example, geologic mapping of the lunar shallow subsurface combines radar transects, which rover-based ground-penetrating radar has used to image the subsurface to depths of about 40 m [6], with measurements at slopes, pit rims, and pit floors [7], [8] that a wheeled rover cannot reach, so that the task falls to a team in which wheeled, legged, and hopping robots each cover the terrain suited to them. Multi-robot autonomy has been demonstrated on the ground for a team of identical wheeled rovers in the CADRE [9] lunar technology demonstration. A team of mixed locomotion modes requires, in addition, mobility and control on loose regolith for each mode, planning under uncertainty, fault detection and recovery, and state estimation under lunar illumination. Each of these functions depends on wheel–terrain or foot–terrain contact and on illumination, which are difficult to model, and they have to be developed and validated on a physical testbed before they can be flown.

Historically, regolith test beds have been built by several laboratories for wheel–soil, mobility, and operations testing; a selection of these facilities is shown in the following Table 1. Existing facilities can be classified by their primary purpose into terramechanics beds for single-wheel or single-vehicle traction testing on prepared and tilted surfaces [10], [11], [12], large analog halls for astronaut training, operations rehearsal, and in-situ resource utilization [13], [14], [15], and yards and sandboxes [16], [17], [18], [19] for rover mobility and navigation development. Motion-capture ground truth is available in several of the rover-oriented facilities [17], [20], [19], [10], and controlled low-angle illumination has been

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emphasized in particular by the LunaLab at the University of Luxembourg [18], [20], which uses a single high-power lamp on a ceiling rail to reproduce polar illumination. The present facility is designed to evaluate wheeled, legged, and hopping platforms under a common terrain, lighting, and motion-capture configuration.

In this paper, we describe the development of a new lunar terrain facility, the Lunar Terrain Facility at the Georgia Tech AstroX [23], that is designed to be robot-centric and to accommodate wheeled, legged, and hopping platforms and multi-robot experiments with minimal reconfiguration. The facility was designed with a 16-camera motion-capture system in view of the goal of testing collaborative autonomy with up to three robots in the bed at once, and with four independently positioned high-output Aputure LS 600d lampswhose position, orientation, beam angle, and dimming are set per experiment. The terrain is a low-fidelity physical analog, and the illumination is adjustable and directional. The experiments evaluate autonomy on coarse granular terrain under adjustable lighting; they do not predict performance on lunar regolith or the appearance of lunar images. In Section 2, we discuss the design rationale.

We developed NOVA as the primary rover platform for these experiments. NOVA is a four-wheel rover with independent steering and drive on each wheel. A controller can therefore command a full planar body twist, including lateral velocity, and the steering strategy itself becomes an experimental variable that changes the slip signature on loose material. We derive a kinematic model of the rover, model the actuation pipeline as a first-order lag with an additive residual that collects the terrain effects the nominal model does not explain, and present a tracking control law that compensates the residual with a Lipschitz-bounded learned correction. The kinematic outer loop of this law runs on NOVA in a ROS 2 based software framework [24]; the learned residual has been trained and closed in the loop on the Leo Rover.

The contributions of this paper are as follows: a) the hardware description of the Lunar Terrain Facility, including the terrain bed and simulant, the directional illumination system, and the motion-capture ground-truth system; b) the hardware and software description of the NOVA rover and of the other platforms exercised in the facility; c) the kinematic and actuation model of NOVA, the residual-compensated tracking control law, and its stability analysis; and d) a brief report of the experiments conducted in the facility to date on NOVA tracking, learning-based control, risk-aware navigation, fault estimation, multi-modal mobility, and visual-inertial estimation under directional lighting.

The remainder of the paper is organized as follows. Section 2 describes the hardware of the facility; Section 3 describes the hardware and software of the NOVA rover and the other platforms; Section 4 presents the dynamics and control of NOVA; Section 5 describes the experiments conducted in the facility; Section 6 states the scope and limitations of the facility and the planned characterization; and Section 7 concludes the paper.

2. The Lunar Terrain Facility at Georgia Tech

Overview of the Facility

The Lunar Terrain Facility (referred to as the Planetary Robotics Facility's basalt simulant pit in [25] and as the lunar regolith simulant lab in [26]) requires the following four components to be operational: the terrain bed, the directional illumination system, the motion-capture system, and the ground station. The terrain bed is an indoor rectangular enclosure of $19\ \mathrm{ft} \times 13\ \mathrm{ft}$ ($5.8\ \mathrm{m} \times 4.0\ \mathrm{m}$, approximately $23\ \mathrm{m}^2$) filled to a nominal depth of 12 in (0.30 m), a nominal volume of approximately $7.0\ \mathrm{m}^3$, with 7,000 kg of basalt gravel from the Merriam Crater deposit in Arizona, the source material of NASA's JSC-1 and JSC-1A lunar simulants. Figure 1 shows the bed with rock obstacles under single-lamp illumination, with the rover ruts and the hard shadow boundaries that the lighting produces. The second component, the illumination system, consists of four Aputure LS 600d daylight-balanced LED fixtures fitted with F10 Fresnel lenses and mounted on stands around the perimeter of the bed. The third component, the motion-capture system, consists of 16 Vicon cameras mounted around the bed that return the six-degree-of-freedom pose of every rigid body carrying a retroreflective marker set at 100 Hz. The ground station runs the experiment control, the Vicon bridge, and the data logger, and communicates with the robots over Wi-Fi. Table 2 lists the current configuration of the facility.

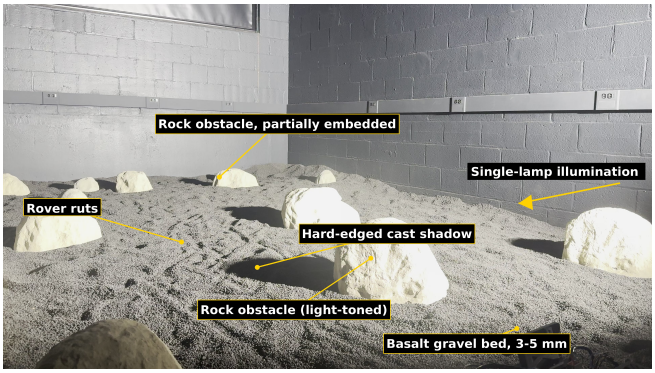


Figure 1: The Lunar Terrain Facility under single-lamp directional illumination from an LS 600d with F10 Fresnel, showing the basalt gravel bed, light-toned rock obstacles placed on and partially embedded in the bed, rover ruts, and the hard-edged shadows cast away from the lamp.

Design Rationale

The facility was designed for the development of autonomy algorithms at technology readiness levels (TRL) 1–4 [30], at which an algorithm is taken from concept to proof of concept and laboratory validation, and not for flight terramechanics or hardware qualification, which require a simulant with the particle-size distribution and cohesion of lunar regolith, thermal-vacuum conditions, and eventually reduced gravity. The facility provides reconfigurable terrain that produces measurable slip and sinkage, adjustable illumination, an independent pose reference, and a bed that can be reset between runs.

The learning-based and robust-control algorithms that the facility serves treat the terrain as unknown. The residual-learning controller of [31] learns the deviation from a rigid-ground model from proprioception and is evaluated on slopes it has not seen; the risk-aware planner of [32] bounds the residual-model error with conformal prediction and plans against the bound rather than against a soil model; and the fault estimator of [25] treats slip and faults as biases on the body twist that are estimated from the inconsistency between proprioception and an exteroceptive anchor. These algo-

Table 1: Planetary terrain facilities at other institutions, with the Lunar Terrain Facility in the last row. Entries marked – are not published.

Organization	Facility	Bed / area	Material	Ground truth	Illumination
ESA / DLR [13], [21]	LUNA, Cologne	700 m^2; 3 m deep drilling area	EAC-1A [22], ≈900 t	–	Sun simulator
Univ. of Luxembourg [18], [20]	LunaLab	11×7 m	Basalt gravel, ≈20 t	Mocap, 12 cam.	1 lamp on rail
NASA Glenn [10]	SLOPE	$6 \times 5 \times 0.3$ m tilt bed, to 45°	GRC-1	OptiTrack, 32 cam.	–
NASA Kennedy [15]	GMRO regolith bin	$7.6 \times 7.6 \times 1.1$ m	BP-1, ≈120 t	–	–
NASA JPL [16]	Mars Yard III (outdoor)	66×36 m	Sand, granite, cinders	–	Sun
ESA ESTEC [17]	Planetary Robotics Lab	9×9 m sandbox	Sand, gravel, rock	Vicon	–
DFKI Bremen [19]	Multifunctional Hall	550 m^2, 105 m^2 crater	Fine granulate	Vicon, 21 cam.	–
NASA Ames [12]	Regolith Testbed	$4\times4\times0.5$ m; $19\times4\times0.3$ m	JSC-1A; LHS-1	–	–
Georgia Tech (this work)	Lunar Terrain Facility	$5.8 \times 4.0 \times 0.3$ m	Merriam Crater basalt gravel, 3–5 mm, 7 t	Vicon, 16 cameras	4 Aputure LS 600d, 15–45° beam

Table 2: Configuration of the Lunar Terrain Facility.

Element	Current configuration
Terrain area	19 ft×13 ft ($5.8\,\mathrm{m}\times4.0\,\mathrm{m}$, $\approx 23\,\mathrm{m}^2$)
Fill depth	12 in (0.30 m), measured
Granular material	Merriam Crater basalt gravel (JSC-1 source deposit), Space Resource Technologies [27]
Particle size	3–5 mm pebbles
Simulant mass / volume	7,000 kg; nominal fill volume ≈ $7.0\,\mathrm{m}^3$
Terrain features	20 loose rock obstacles, 10–40 cm; slopes of 10–30°, berms, and depressions shaped in the bed
Lighting	Four Aputure LS 600d daylight (5600 K) fixtures with F10 Fresnel lenses
Ground truth	Vicon motion capture, 16 cameras (10 Vantage V5 [28], 6 Vero v2.2 [29]), Vicon Tracker 3.10, 100 Hz, ≈2 mm static accuracy [25], mean calibration world error 0.43 mm (max 0.56 mm), 6×6 m tracked floor area
Platforms	NOVA (4-wheel independent steer); Leo Rover (skid steer); Unitree G1 humanoid; DUALITY wheel-leg rover; Multimodal Mini legged rover; hopping platforms (in development)

rithms require a terrain that produces sinkage and slip that depend on load, steering, and slope, with enough variability between preparation states to test generalization. The coarse basalt gravel provides this at a fraction of the cost of a fine-grained simulant. It can be raked flat and re-dressed in minutes without dust control, it has shown no visible segregation or compaction over months of use, although its mechanical properties have not been measured over that period, and it presents a visually uniform grey surface without the color and texture cues of terrestrial soils. Traction questions for specific wheel or foot designs are referred to the terramechanics facilities of Table 1.

The perception algorithm responds to illumination changes that cannot be compensated downstream at controller level.

Table 3: Manufacturer-reported center-beam photometrics for one LS 600d with F10 Fresnel [33].

	1 m	3 m	5 m
45° flood	89,200 lx	11,440 lx	4,260 lx
15° spot	224,200 lx	29,300 lx	10,580 lx

For example, a stereo matcher that loses texture in a saturated or shadowed region returns no depth there, and a segmentation model trained under terrestrial illumination has not seen hard-edged shadows several body lengths long across a surface of uniform albedo. Lunar polar imagery differs from terrestrial training data in this respect, and datasets built to study the resulting distribution shift have reproduced low-angle single-source lighting physically [3], [4]. The facility therefore uses adjustable lamps whose position, elevation, azimuth, beam angle, and dimming are set and recorded, and treats the resulting shadow geometry and contrast as a controlled stressor on perception rather than as a reproduction of lunar illumination; the source subtends a larger angle at the target than the Sun does, its spectrum is that of a 5600 K LED.

The quantities evaluated in the experiments (estimation error, slip, tracking error) are differences between what the robot measures or commands and its motion, which requires a pose reference that is independent of the autonomy under test. The motion-capture system supplies this reference, and it allows a run to be repeated with the controller closed on the motion-capture pose and then on an onboard estimate, which separates the two feedback sources when a failure is investigated. Finally, the experiments are many short runs with one variable changed between them, which favors a bed that can be raked flat and re-dressed with rocks and relief in minutes over a fixed scene.

Directional Illumination System

The facility uses four Aputure LS 600d daylight-balanced LED fixtures fitted with Aputure F10 Fresnel lenses. Each fixture provides adjustable dimming (0–100%, 5600 K) [33] and beam angle (15°–45°) [34]. Lamp position, orientation, beam angle, and dimming define the experimental lighting configuration. The photometric values in Table 3 are manufacturer specifications for one LS 600d with the F10 Fresnel [33], not measurements of the installed facility.

For a nominal full beam angle θ and source distance d,

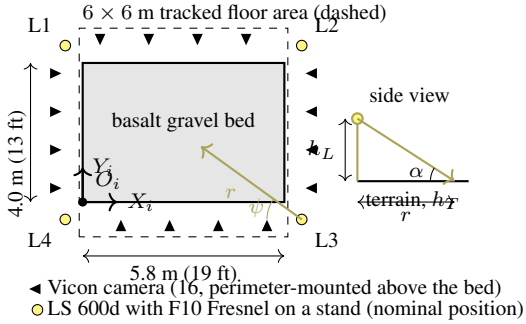


Figure 2: Layout of the illumination and motion-capture systems around the terrain bed, drawn to scale for the bed and the tracked floor area. The four LS 600d lamps are on movable stands, so their positions are nominal; for each run the lamp position, height h_L, azimuth ψ, elevation α of Eq. (1), beam angle, and dimming level are surveyed in the motion-capture frame and recorded. The 16 cameras are mounted around the perimeter above the bed.

the geometric beam diameter is approximately $D(d, \theta) = 2d \tan(\theta/2)$, which gives 0.79 m for a 15° beam and 2.49 m for a 45° beam at $d = 3$ m, and 1.32 m and 4.14 m, respectively, at $d = 5$ m. Since the lamps are placed around the perimeter of the $5.8\,\mathrm{m} \times 4.0\,\mathrm{m}$ bed, source-to-target distances fall in the range of roughly 2–7 m depending on the aim point; a single lamp in the flood setting then covers the full bed width, while the spot setting confines illumination to a region comparable to the rover footprint. The illuminance at the bed from one or more lamps depends on aiming, beam overlap, dimming, surface orientation, and spatial nonuniformity and must be measured in the installed configuration.

The illumination geometry is summarized by the source elevation angle

$$\alpha = \tan^{-1}\left(\frac{h_L - h_T}{r}\right), \tag{1}$$

where h_L is the source height, h_T is the local terrain reference height, and r is the horizontal source-to-test-region distance, as shown in Fig. 2. Low source elevation generates elongated shadows and strong lit-to-shadow transitions, a change in azimuth moves the shadows relative to a robot trajectory, and beam angle and dimming vary the illuminated footprint and intensity without changing the terrain geometry; Fig. 4 shows the resulting hard shadow edges and saturated lit rock faces when a single lamp is used with the remaining three lamps off. Since the lamp positions are surveyed in the motion-capture world frame, the shadow geometry can be predicted from the ground-truth terrain and obstacle poses, and since the motion-capture pose reference is available under every lighting configuration, the effect of illumination on the perception and estimation layers can be examined separately from its effect on the downstream planner and controller.

Terrain Bed and Simulant

JSC-1 is a basaltic simulant originally developed for large- and medium-scale lunar engineering studies from volcanic ash quarried at Merriam Crater in the San Francisco volcanic field of northern Arizona, and its successor JSC-1A has been characterized in terms of composition and geotechnical behavior [35], [36]; NASA guidance emphasizes that no single simulant reproduces all properties of lunar regolith and that simulant selection should be driven by the intended test objective [37]. The facility bed is filled with 7,000 kg of basalt gravel from the same Merriam Crater deposit, supplied by Space Resource Technologies in 1 kg units as pebbles of approximately 3–5 mm, so that the material comes from the same source deposit as the JSC-1 family and as the vendor's engineering-grade successor LMS-1E [27], while being more than an order of magnitude coarser than the sub-millimeter grain sizes of those simulants; a common source deposit does not by itself establish a common composition or mechanical behavior, and the gravel has not been characterized in either respect. In use it behaves as a free-flowing, low-cohesion granular medium with a visually uniform grey surface. The fill depth, measured at 0.30 m in the installed bed, is several times the sinkage observed for the platforms used to date, so that the wheels and feet of a small robot interact with the gravel rather than with the rigid floor beneath the bed. The plan dimensions support straight-line traverses of up to approximately 5 m and curved, obstacle-avoidance, and multi-robot paths of comparable length while keeping the entire bed accessible for terrain reset and sensor calibration from the perimeter.

For the robotics experiments conducted in the facility, the coarse granular material serves three roles: it introduces non-rigid wheel–terrain and foot–terrain interaction with visible sinkage and measurable slip that can be compared against the motion-capture ground truth; it can be reshaped by hand to create local slopes, depressions, and berms and raked flat to reset the bed between runs; and its granular texture, ruts, footprints, and relief shadows provide visual and geometric structure for perception and mapping experiments, while its visually uniform grey surface removes the color and texture cues that terrestrial soils would otherwise provide to a learned perception model. The particle size is intentionally coarser than the fine fraction present in lunar regolith, and the gravel has not been geotechnically characterized as JSC-1A and LMS-1 have [36], [38], so that experiments in the facility are not interpreted as direct predictions of lunar terramechanics; the wheel sizing of the DUALITY rover [26], for instance, was carried out with published parameters of the finer LMS-1E product rather than of the gravel itself.

Two classes of terrain features are used. The first is a set of loose, light-toned rock obstacles of varying size that are placed on or partially embedded in the bed, which allows obstacle density, spacing, and path width to be changed between runs without disturbing the rest of the surface; their high albedo relative to the grey simulant produces strong lit-face to shadow contrast under directional lighting, which is visible in Fig. 1, and they can be fitted with retroreflective markers, in which case their poses are returned by the motion-capture system as a ground-truth obstacle map. The second is relief shaped by hand directly in the simulant, with no substrate beneath the gravel, so that its geometry is bounded by the fill depth and the angle of repose of the material; slopes of 10° to 30° have been built for the wheel-leg mobility tests, and a sloped portion of the bed was used to generate out-of-distribution slope data for the residual-learning controller. Each terrain configuration is recorded together with the run in terms of longitudinal and cross-slope, obstacle height, obstacle density, path width, local roughness, and preparation state.

Motion-Capture Ground Truth

The pose of every robot and marker-equipped object in the bed is measured using 16 Vicon motion-capture cameras (10 Vantage V5 [28] and 6 Vero v2.2 [29]) mounted around the perimeter of the facility, which return rigid-body poses at

100 Hz with a static accuracy of approximately 2 mm, as used in the fault-estimation experiments of [25]. The sixteen cameras provide more views than a single robot on a flat floor would require, which is of use in the facility in three ways: the rock obstacles and the shaped relief occlude low-mounted markers from part of the perimeter; a camera looking toward a lamp at low elevation can be blinded, so that the views from the opposite side carry the reconstruction; and the planned multi-robot experiments will place several small marker sets in the volume at once. The motion-capture pose reference is used in three ways: as the reference against which onboard state estimators and visual-inertial odometry are evaluated as lighting, terrain, and slip are varied; as the feedback to the controller, or as a low-rate exteroceptive anchor in place of a sun sensor or GPS, in experiments whose purpose is to examine planning, control, or fault-detection behavior with the estimation error removed from the loop; and as the poses of marker-equipped obstacles, which yields a reference map for evaluating onboard mapping. The 100 Hz Vicon twist also serves as the supervision target for learned residual dynamics, and the 4 Hz subsampled Vicon pose serves as the exteroceptive anchor for the invariant observer.

Onboard sensor streams, state estimates, commanded actions, controller and planner status, and health events are logged on the robot, while the motion-capture poses and the lighting and terrain configuration are logged at the ground station, and the two logs are aligned on a common time base so that the motion-capture pose is associated with each onboard measurement and command. A single run can therefore be replayed against a modified estimator, planner, or controller offline, and slip is computed from the difference between commanded, odometric, and motion-capture motion.

The cameras are ceiling-mounted, cover a 6×6 m tracked floor area, and are operated with Vicon Tracker 3.10; the calibration used for the experiments reported here yielded a mean per-camera world error of 0.43 mm (maximum 0.56 mm) and a mean image reprojection error of 0.24 px, and the 2 mm static accuracy is the figure returned by the dynamic calibration routine with the Active Wand [39]. The rover pose is streamed to its onboard computer through the `vicon_receiver` ROS 2 bridge, which carries the native Vicon frame-acquisition timestamps so that the poses are aligned to the onboard clock by timestamp rather than by arrival time.

3. Robotic Platforms and Software

NOVA is the primary platform developed for the facility and is shown on the bed in Figs. 3 and 4; Fig. 3 labels the four steering modules, the drive motors and belts, the stereo camera, and the compute enclosure that the following subsections describe. Each wheel is independently steered and driven, and the rover can therefore execute point turns, crab motion, and Ackermann-like arcs on the granular surface without the wheel scrub that a skid-steer platform imposes, while a controller can command a full planar body twist including lateral velocity. The hardware on the rover is divided into three subsystems: 1) mechanical, including the chassis, suspension, and wheel modules; 2) electrical, including power, computing, and low-level controller boards; and 3) sensing, including the exteroceptive and proprioceptive sensors and the motion-capture marker set. Table 4 lists the components of each subsystem, and Table 5 summarizes the configuration.

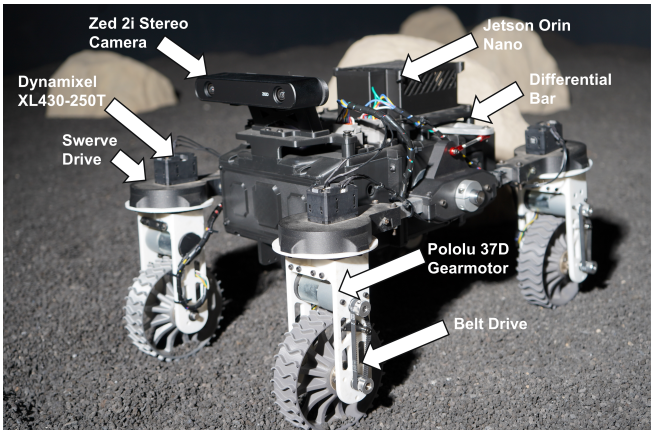


Figure 3: Annotated view of the NOVA rover on a granular test bed, highlighting its four independently steered wheel modules, drivetrain components, stereo camera, and chassis-mounted computing enclosure.

Figure 4: NOVA traversing between rock obstacles under single-lamp illumination. The lit rock faces, hard shadow boundaries, and ruts are the appearance and terrain cues that the perception and estimation layers must handle, while the motion-capture system provides an independent pose reference.

Mechanical Subsystem

The chassis is built from 3D-printed PLA structures and $15 \times 15 \times 2$ mm, 300 mm long 6063 aluminum square tube members, with brass heat-set inserts at all bolted joints, and the left and right wheel pairs are joined through a passive differential bar with ball-joint linkages, which allows the wheel pairs to conform to local terrain relief and keeps all four wheels loaded on uneven ground. Each swerve module is a two-stage assembly: an upper steering stage in which a Dynamixel XL430-W250-T servo [40] rotates the module about a vertical axis on a 65 mm turntable bearing, and a lower drive stage in which a Pololu 100:1 metal gearmotor (37D$\times$73 mm 64 CPR encoder) drives an 8 mm wheel shaft through a 220 mm GT2 timing belt and 20-tooth pulleys, so that the drive motor is carried above the wheel and the belt run spans the steering axis. This arrangement keeps the module compact and places the motor mass close to the chassis, at the cost of a belt compliance between the motor and the wheel that the current open-loop drive interface of Section 4 does not compensate. The swerve architecture is shown as an exploded view in Fig. 5. The differential bar and the belt-drive module are the two mechanical features that distinguish NOVA from a skid-steer platform of the same size and that

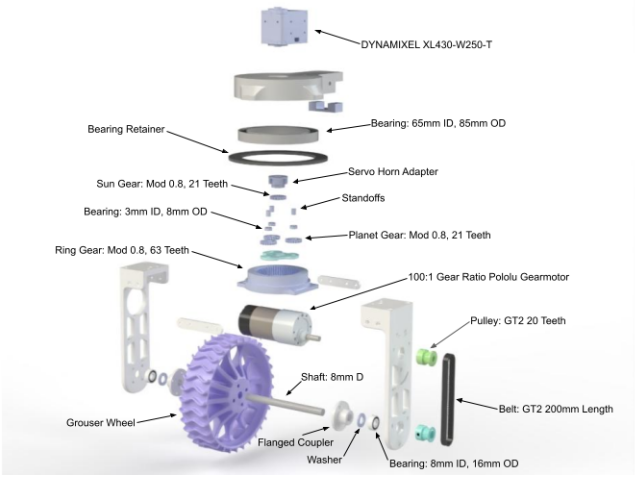


Figure 5: Exploded view of a NOVA swerve-drive module, showing the servo-driven planetary steering mechanism, gearmotor-driven timing-belt transmission, grouser wheel, and supporting bearings, shaft, and structural components.

make its slip signature on loose material a function of the commanded steering.

Electrical Subsystem

The electrical architecture of NOVA is shown in Fig. 6. Power is supplied by a 3S 11.1 V, 10 Ah lithium-polymer battery through a Matek power distribution board that provides regulated 5 V, 9 V, and 12 V rails, with a lighted rocker switch and EC5 connectors on the battery side and Deutsch DTM sealed connectors between the chassis and the wheel modules. We chose an NVIDIA Jetson Orin Nano Super developer kit (8 GB, 67 TOPS) [41] as the main computer to run the perception, estimation, planning, and control algorithms, and a Raspberry Pi Pico 2W microcontroller as the low-level controller that receives signed duty-cycle commands for each wheel from the Jetson and generates the PWM and direction signals for four Cytron MDD10A dual-channel motor drivers; the drive motors carry 64 CPR encoders, but the Pico firmware in use does not yet read them, so that the drive stage runs open loop at the wheel-speed level. The four steering servos are driven over a single Dynamixel bus through a ROBOTIS U2D2 interface and power hub, which returns the servo position, velocity, and load to the Jetson at the bus rate. A 50 mm fan cools the compute enclosure, which is a 3D-printed housing with a filtered intake to keep simulant dust away from the board.

The electrical architecture separates high-level autonomy from low-level, time-critical actuator control: motion commands generated on the Jetson are converted through the rover's actuation layer into steering angles, which are sent over the Dynamixel bus where each servo closes its own position loop, and into drive duty cycles, which the Pico converts to PWM and direction signals without a wheel-speed loop, while the servo feedback and the IMU measurements are returned to the Jetson for state estimation and closed-loop control, which at present closes on the motion-capture pose. This division lets the autonomy algorithms of Sections 4 and 5 interact with the rover through a consistent motion-command interface that isolates them from the motor electronics.

Table 4: List of components on the NOVA rover.

Subsystem	Component	Manufacturer
Mechanical	Chassis structure (PLA, Al tube)	In-house
	Differential bar and linkages	In-house
	Steering module, turntable bearing	In-house / uxcell
	Belt drive (2GT belt, 20T pulleys)	uxcell / WINSINN
	Wheels	In-house
Actuation	Drive gearmotor, 100:1, 37D	Pololu
	Motor driver, MDD10A	Cytron
	Steering servo, XL430-W250-T	ROBOTIS
	Servo interface, U2D2 + hub	ROBOTIS
Electronics	Computer, Jetson Orin Nano 8 GB	NVIDIA
	Low-level controller, Pico 2W	Raspberry Pi
	Power distribution board	Matek
	Battery, 3S 10 Ah LiPo	Zeee
	Connectors, DTM series	Deutsch
Sensing	Stereo camera, ZED 2i	Stereolabs
	IMU, BNO055	Bosch / Adafruit
	Motion-capture markers	Vicon

Sensing Subsystem

Exteroceptive sensing is provided by a Stereolabs ZED 2i stereo camera [42] mounted at the front of the chassis, which provides synchronized stereo images, depth, and its own integrated inertial measurements over USB to the Jetson. The camera is configured at 720p and 60 Hz. Proprioceptive sensing is provided by a Bosch BNO055 nine-axis IMU [43] on the chassis, by the position, velocity, and load of the four steering servos returned over the Dynamixel bus, and by the duty-cycle commands sent to the low-level controller, which together are intended to form the proprioceptive vector (body velocity, IMU specific force and angular rates, steering state, and drive commands) for the learned residual of Section 4 once the drive encoders are read; the residual model has so far been trained only on the Leo Rover (Section 5). The IMU provides inertial measurements at approximately 100 Hz. A retroreflective marker set mounted on the chassis defines the Vicon rigid body used to track the rover pose in the motion-capture system of Section 2 (Fig. 7).

Other Platforms Exercised in the Facility

The facility is equipped to address autonomy research questions for multi-modal and multi-agent robotics for exploration and test them on the same terrain under the same ground truth. Figure 8 shows the platforms that have been run on the bed to date, with panels reproduced from the papers in which the corresponding experiments are reported.

Leo Rover—The Leo Rover [46] is a four-wheel skid-steer differential-drive platform with wheel encoders, an IMU, and a camera, running ROS on an onboard Raspberry Pi

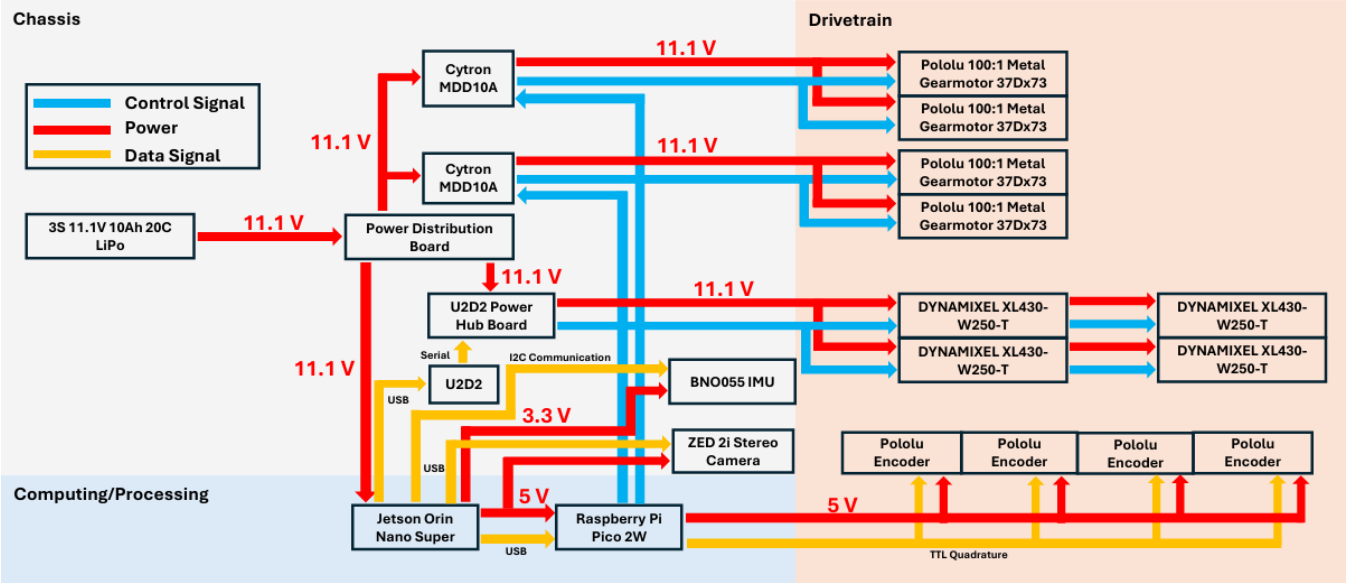


Figure 6: Control block diagram of NOVA, showing power distribution, onboard computing and communication protocols. Red, blue, and yellow lines denote power, control, and data connections, respectively.

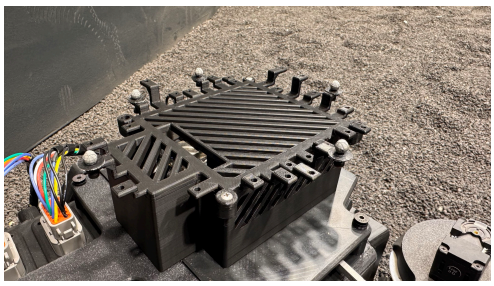


Figure 7: Placement of the retroreflective Vicon markers on the NOVA chassis.

at 30 Hz. It is the platform on which the proprioceptive residual-learning controller of [31] and the invariant adaptive fault estimator of [25] were developed, and, fitted with a ZED 2i stereo camera and an NVIDIA Jetson AGX Orin, it is the platform on which the risk-aware planning and control architecture of [32] was demonstrated. It serves as a contrast to NOVA because its steering produces lateral wheel slip by design, and the same terrain therefore yields a different residual-dynamics signature on the two rovers.

Unitree G1 humanoid— The 35 kg, 29-degree-of-freedom Unitree G1 is the platform on which the terrain-adaptive locomotion policy of [44] was deployed, walking and running on basalt at commanded velocities of 0.5–1.5 m/s and on dry sand at up to 2.5 m/s with a policy trained entirely in simulation against a granular contact model. The humanoid experiments evaluate locomotion on the same gravel bed used for the rover tests.

DUALITY wheel-leg rover— DUALITY [26] is a student-designed hybrid wheel-leg quadruped with a 280 mm × 360 mm 3D-printed chassis, four two-degree-of-freedom legs (hip abduction and a four-bar lifting joint) each terminated by a wheel, twelve GoBilda servos, a Raspberry Pi 5, and a nine-axis IMU, which switches between a rolling mode on flat terrain and a crawling gait on slopes. It is the mode-switching platform on which the 10–30° slope tests of Section 5 were conducted.

Table 5: NOVA rover configuration.

Subsystem	Description
Mobility	Four wheels, each independently steered and driven; passive differential-bar suspension; 130D x 35mm width wheel, grousers 2 x 5mm (width x height)
Dimensions / mass	506 x 371 x 358 mm (length x width x height); 7.4725 kg
Primary compute	NVIDIA Jetson Orin Nano (8 GB); Raspberry Pi Pico 2W for low-level motor interfacing
Cameras	Stereolabs ZED 2i stereo camera: 720p at 60 Hz
Inertial sensing	Bosch BNO055 9-DOF IMU: 100 Hz
Proprioception	Steering servo position, velocity, and load; drive-motor encoders (64 CPR quadrature) fitted but not yet read by the low-level controller
Ground truth	Vicon rigid-body marker set
Networking	Wi-Fi to ground station; ROS 2 Jazzy on Ubuntu 24.04
Power	3S 10 Ah 11.1V 120C LiPo; 28.5min nominal test duration

Legged platforms and adaptive wheels— Legged platforms like the Multimodal Mini rover developed in the lab for traversing rubble and steep relief that a wheeled robot cannot negotiate are exercised on the same bed; the hopping platforms of the mission concept are in development and have not yet been run on the bed. The GROVER variable-height grouser wheel [47], [45], whose grousers extend continuously between 0 and 17.5 mm through a servo-driven spiral cam, was developed and tested on a single-wheel gantry testbed in the lab, not on the facility bed, and is being integrated onto a

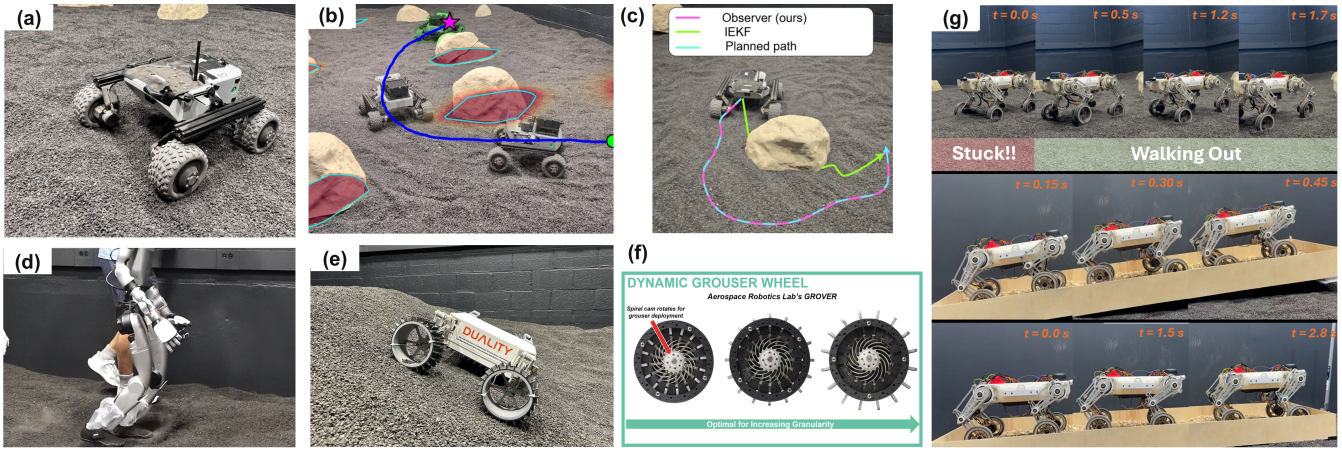


Figure 8: Platforms exercised in the Lunar Terrain Facility, with the exception of (f). (a) The Leo Rover on the basalt gravel bed during the proprioceptive residual-learning data collection, reproduced from [31]. (b) The Leo Rover with ZED 2i and Jetson AGX Orin executing the risk-aware planner between rock obstacles, with the risk map and the tracked trajectory overlaid, reproduced from [32]. (c) The Leo Rover under an injected fault, with the fault-tolerant observer path (magenta), the invariant EKF path (green), and the planned path (cyan), reproduced from [25]. (d) The Unitree G1 humanoid walking on the basalt bed under the terrain-adaptive policy, reproduced from [44]. (e) The DUALITY wheel-leg rover rolling on a slope shaped in the bed, reproduced from [26]. (f) GROVER, a variable-grouser-height wheel, at three grouser heights on the single-wheel gantry testbed in the lab, not on the facility bed, reproduced from [45]. (g) The Multimodal Mini using its legged degrees of freedom to walk across simulated lunar terrain and ascend slopes.

rover for full-vehicle tests in the facility.

Software Architecture

The software for NOVA and the other platforms was designed to allow for interchangeable estimation, mapping, planning, and control modules, so that the same planner and the same estimator can be evaluated on different wheeled platforms, and in principle on legged and hopping platforms, with only the platform-specific dynamics, actuation interface, and learned residual model exchanged at the control layer. As illustrated in Fig. 9, the architecture is organized as follows: the state-estimation module fuses the proprioceptive and stereo measurements, or substitutes the motion-capture pose, and publishes the current state and its covariance; the mapping module converts the stereo point cloud into a distance field whose obstacle boundaries are inflated by the mapping uncertainty, following [32]; the planning module maintains a trajectory of desired states from which the controller selects a target state at each timestep; the control module implements the tracking law of Section 4 and publishes body-twist commands, which the low-level interface converts to steering angles and wheel speeds; and the health module monitors estimator confidence, planner feasibility, controller saturation, and the fault estimator of [25], and publishes intervention events. On NOVA, the setup in use at the time of writing is motion-capture based estimation, a preselected maneuver in place of the planner, and the kinematic outer loop of Section 4; the risk-aware kinodynamic planner of [32] and the residual-compensated inner loop have been run on the Leo Rover in this architecture and their transfer to NOVA is in progress. The modules can be swapped for onboard visual-inertial estimation, an arbitrary motion planner, or a learned locomotion policy in the case of the legged platforms.

The architecture is backed by ROS 2 [24], which allows each module to be scheduled at its own rate that can be changed at run time, provides the messaging layer between the onboard computer, the low-level controller, and the ground station, and provides launch configurations for the different platform and estimator-source setups described in Section 2. In the facility, the estimator source (onboard or motion capture) is a launch-time parameter, and the same run can be repeated first under motion-capture feedback and then under onboard estimation with no other change.

The Jetson onboard runs ROS 2 Jazzy on Ubuntu 24.04. The baseline trajectory-tracking node executes at 20 Hz, polls the Pico serial interface at 100 Hz and publishes operating status at 1 Hz. Vicon poses are received on `/vicon/nova/nova`.

The operator selects the maneuver before launch and starts or stops the run through the ROS services `/nova/start` and `/nova/stop`. The wheel modules are manually aligned forward before launch and the measured startup positions define the steering references. Driving is disabled until the start request is accepted. The controller requires one second of valid Vicon data and monitors pose validity, control-loop delays, run duration and tracking error. A stop condition disables driving.

4. Dynamics and Control of NOVA Rover

In this section, we present the kinematic and actuation model of NOVA, the tracking controller, its stability analysis, and the implementation on the rover. The controller is a cascade: an outer kinematic loop maps the configuration error to a reference body twist, an inner loop drives the realized twist to that reference through a first-order actuation model with a learned terrain residual, and an actuation layer maps the commanded twist to the eight actuator set points. The coordinate systems are shown in Fig. 10. The world frame (X_i, Y_i, Z_i) with origin O_i is the motion-capture frame fixed to the bed, and the body frame (X_b, Y_b, Z_b) with origin O_b is attached to the chassis at the geometric center of the four steering axes, with X_b pointing forward and Z_b up.

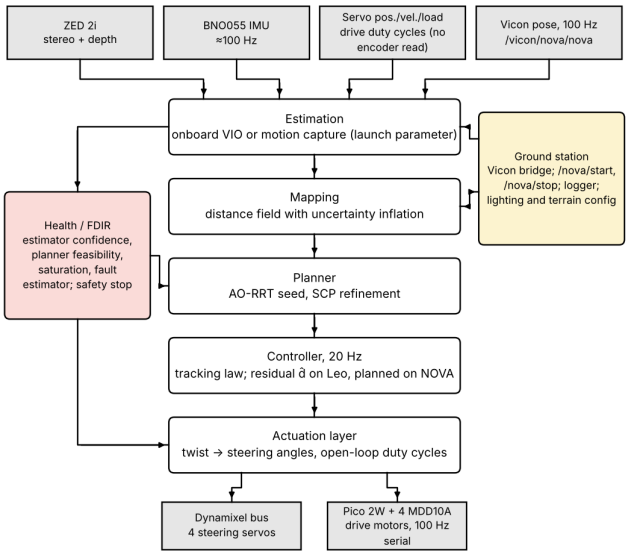


Figure 9: Software architecture of the autonomy stack. Sensor inputs (top) feed the estimation and mapping modules; the planner, controller, and actuation layer form the command path; the health module can trigger a replan or a safety stop; and the ground station provides the motion-capture pose, run control, and logging. Each module is a ROS 2 node with its own rate, and the estimator source is a launch-time parameter. On NOVA the modules run to date are motion-capture estimation, a preselected maneuver, the kinematic outer loop, and the actuation layer; the mapping, planning, and residual modules have been run on the Leo Rover.

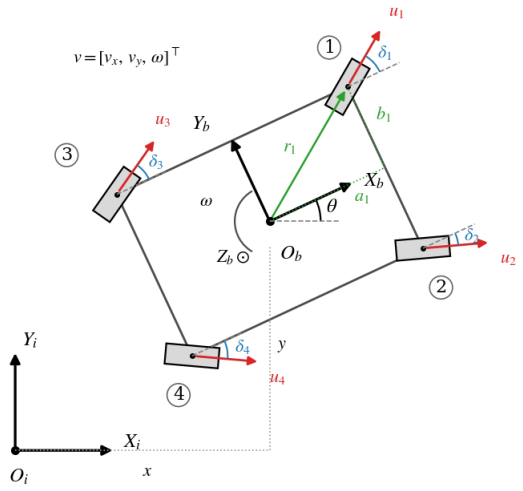


Figure 10: Coordinate systems and wheel nomenclature used for the derivation of the kinematic model. The body frame (X_b, Y_b, Z_b) at O_b on the NOVA chassis, with the wheel indices $i = 1, \ldots, 4$. Plan view with the world frame (X_i, Y_i) at O_i, the configuration (x, y, θ), the body twist $v = [v_x, v_y, \omega]^\top$, the wheel positions $r_i = (a_i, b_i)$, the steering angles δ_i measured from X_b, and the wheel-center velocities u_i.

Model

Configuration and wheel kinematics—Let $q = [x,\ y,\ \theta]^\top$ denote the planar configuration of the rover in the world frame and $v = [v_x,\ v_y,\ \omega]^\top$ the body twist, i.e., the longitudinal and lateral velocity of O_b expressed in the body frame and the yaw rate. The configuration kinematics are

$$\dot{q} = S(q)\, v, \qquad S(q) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

where $S(q)$ is orthogonal. Let $r_i = [a_i,\ b_i]^\top$, $i = 1, \ldots, 4$, denote the position of the steering axis of wheel i in the body frame. Under the no-slip assumption, the velocity of the wheel center of wheel i in the body frame is

$$u_i = \begin{bmatrix} v_x - \omega\, b_i \\ v_y + \omega\, a_i \end{bmatrix} = J_i\, v, \qquad J_i = \begin{bmatrix} 1 & 0 & -b_i \\ 0 & 1 & a_i \end{bmatrix}, \tag{3}$$

and the steering angle δ_i and the wheel angular rate $\dot{\phi}_i$ that realize this velocity with a wheel of radius R_w are

$$\delta_i = \operatorname{atan2}(u_{i,y},\ u_{i,x}), \qquad \dot{\phi}_i = \frac{\|u_i\|}{R_w}. \tag{4}$$

Equation (4) is the inverse kinematics: a twist in $\mathbb{R}^3$ fixes all eight set points, and δ_i is defined modulo π with a corresponding sign change of $\dot{\phi}_i$. In the other direction the problem is overdetermined. Stacking the measured wheel-center velocities $\bar{u}_i = R_w \dot{\phi}_i\, [\cos\delta_i,\ \sin\delta_i]^\top$ into $\bar{u} \in \mathbb{R}^8$ and the J_i into $J \in \mathbb{R}^{8\times 3}$, which has full column rank for any four non-collinear wheel positions, we compute the odometric twist as the weighted least-squares solution

$$\hat{v} = (J^\top W J)^{-1} J^\top W \bar{u}, \qquad W = \operatorname{diag}(w_1 I_2, \ldots, w_4 I_2), \tag{5}$$

where the per-wheel residual $\varepsilon_i = \|\bar{u}_i - J_i \hat{v}\|$ measures how far wheel i departs from rolling consistently with the other three, and the weights $w_i = 1/(1 + \gamma\varepsilon_i)$, $\gamma > 0$, discount a slipping wheel in the odometry. On rigid ground ε_i is at the encoder noise level and $W = I$; on regolith ε_i carries the slip signature of the wheel.

Operating modes— NOVA is operated in two modes that share the chassis, the drive motors, and the software stack and differ only in the actuation layer. In *swerve mode* all four steering servos are active and the full twist $v \in \mathbb{R}^3$ is realized through (4); the rover can translate in any body direction, execute point turns about O_b, and follow arcs with the wheels aligned to the local velocity, so that the heading and the direction of travel are independent inputs. In *differential-drive mode* the steering servos are held at $\delta_i = 0$ and the four drive motors are commanded as a left pair ($b_i > 0$) and a right pair ($b_i < 0$). The realizable twist is $\bar{v} = [v_x,\ \omega]^\top$, the lateral channel is removed from (2) by the nonholonomic constraint $v_y = 0$, and the wheel speeds follow from the longitudinal component of (3),

$$\dot{\phi}_i = \frac{v_x - \omega\, b_i}{R_w}, \qquad i = 1, \ldots, 4, \tag{6}$$

while the lateral component ωa_i of each wheel-center velocity is absorbed by lateral slip of the wheels during a turn, as on a skid-steer platform. The model and controller below are written for swerve mode; the differential-drive versions are obtained by deleting the lateral channel.

Twist dynamics and residual—The steering servos close their position loops internally, and the drive stage maps a duty-cycle command to wheel torque through the motor driver, the gearmotor, and the belt without a wheel-speed loop (Section 4). Following [31], we model the pipeline from the commanded twist $u \in \mathbb{R}^3$ to the realized twist v as a first-order lag per channel,

$$\dot{v} = A_n v + B_n u + d(t), \tag{7}$$

with $A_n = -\operatorname{diag}(1/\tau_v,\, 1/\tau_v,\, 1/\tau_\omega)$ and $B_n = \operatorname{diag}(k_v/\tau_v,\, k_v/\tau_v,\, k_\omega/\tau_\omega)$. The translational channels share one time constant and one gain because every wheel is steered to the direction of the commanded velocity, while the yaw channel is distinct because a yaw command sets the wheels tangent to a circle about O_b and loads them differently. In differential-drive mode the same parameters act on $\bar{v}$ through $\bar{A}_n = -\operatorname{diag}(1/\tau_v, 1/\tau_\omega)$ and $\bar{B}_n = \operatorname{diag}(k_v/\tau_v, k_\omega/\tau_\omega)$. The steering servo adds a faster stage between the commanded and the realized wheel direction, which we model as a rate-limited first-order lag

$$\dot{\delta}_i = \operatorname{sat}_{\dot{\delta}_{\max}}\left(\frac{\delta_i^c - \delta_i}{\tau_\delta}\right), \tag{8}$$

with δ_i^c the commanded angle from (4) and $\tau_\delta \ll \tau_v$, so that its effect on the twist response is absorbed into the identified τ_v except during large steering transients.

The residual $d(t)$ collects everything that the nominal first-order model does not explain: unmodeled actuator and drive-stage dynamics, belt compliance, the steering-servo transient of (8), the zero-order hold of the 20 Hz command, gravity projection on slopes, the wheel–terrain interaction on deformable regolith (slip, sinkage, and terrain load), and, in differential-drive mode, the lateral scrub of (6). The residual combines unmodeled actuation effects and wheel–terrain interaction and is non-Markovian, since shear strain accumulates and wheels recover from sinkage over several timesteps. The learned residual approximates d as a whole.

Control

Outer loop—The objective is for $q(t)$ to track a trajectory $q_d(t) \in \mathcal{C}^2$ generated by the planner. In swerve mode the rover is holonomic in the plane, and we use the kinematic controller

$$v_r = S(q)^\top\left(\dot{q}_d + \Lambda\,(q_d - q)\right), \qquad \Lambda \succ 0 \text{ diagonal}, \tag{9}$$

which, with $v = v_r + s$ substituted into (2), gives the configuration error dynamics

$$\dot{e} = -\Lambda e - S(q)\,s, \qquad e = q_d - q, \tag{10}$$

where $s = v - v_r$ is the inner-loop error. With $s \equiv 0$, (10) is an exponentially stable tracking loop, and (9) is the controller we use alone when the twist is treated as the control input, which is the case on NOVA at present. In differential-drive mode $v_y = 0$ and the reference $\bar{v}_r = [v_{x,r},\, \omega_r]^\top$ is generated by the nonholonomic tracking controller of [31], whose error dynamics $\dot{e} = g(e,t) - \bar{S}(q)\,s$, with $\bar{S}(q)$ the first and third columns of $S(q)$, are exponentially stable at $s = 0$.

Inner loop—Given the reference twist $v_r(t)$, define $s = v - v_r$ and the control law

$$u = B_n^{-1}\Big(\dot{v}_r - A_n v - \hat{d}(z_t) - Ks\Big), \qquad K \succ 0, \tag{11}$$

which substituted into (7) gives

$$\dot{s} = -Ks + \tilde{d}(t), \qquad \tilde{d} = d - \hat{d}(z_t). \tag{12}$$

The inner loop is a linear system driven by the residual-model error, and the learned residual $\hat{d}(z_t)$ acts as a feedforward cancellation of the part of d that the model predicts from the proprioceptive history z_t. The actuation layer realizes u through (4) in swerve mode or (6) in differential-drive mode, selecting for each servo the branch of δ_i that minimizes the steering travel.

Force-level allocation— The eight actuators are redundant over the three twist components at the force level. Let $F = [F_x,\, F_y,\, M_z]^\top$ denote a generalized force on the body and $f = [f_1, \ldots, f_4]^\top$ the tractive forces along the wheel headings. Then $F = G(\delta) f$ with

$$G(\delta) = \begin{bmatrix} \cos\delta_1 & \cdots & \cos\delta_4 \\ \sin\delta_1 & \cdots & \sin\delta_4 \\ a_1\sin\delta_1 - b_1\cos\delta_1 & \cdots & a_4\sin\delta_4 - b_4\cos\delta_4 \end{bmatrix}, \tag{13}$$

which has full row rank if and only if the lines of action of the four wheel forces are neither all parallel nor all concurrent, and the weighted minimum-norm allocation

$$\begin{gathered} f = W_f^{-1} G^\top \left(G\, W_f^{-1} G^\top\right)^{-1} F, \\ W_f = \operatorname{diag}(1+\gamma\varepsilon_1, \ldots, 1+\gamma\varepsilon_4), \end{gathered} \tag{14}$$

shifts tractive force away from a slipping wheel using the residual of (5) and the weighted generalized pseudo-inverse of [5]. This allocation requires a torque interface to the drive stage; it is not implemented on NOVA, whose current interface sends the kinematic commands of (4) to the wheels.

Stability

We analyze the cascade under the following assumptions.

- A1) $q_d \in \mathcal{C}^2$ with $\dot{q}_d$ and $\ddot{q}_d$ bounded.
- A2) $k_v, k_\omega > 0$, so that B_n is invertible.
- A3) $\|d(t)\| \le D$, and the proprioceptive history z_t remains in a compact set $\mathcal{Z}$ on which the Lipschitz learned residual [48] satisfies $\|\hat{d}(z)\| \le \hat{D}$. Hence $\|\tilde{d}(t)\| \le \bar{d}$ for some $\bar{d} \le D + \hat{D}$.
- A4) The duty-cycle commands do not saturate, the controller is closed on the motion-capture pose with its 2 mm error neglected, and the steering transient (8) and the 20 Hz zero-order hold are included in $d(t)$ and therefore in D.

Under A4 the result is conditional on the bound D and is not a sampled-data proof. Without the learned correction, $\hat{d} \equiv 0$ and $\bar{d} = D$; a value $\bar{d} < D$ is a property of the trained model on $\mathcal{Z}$ and is not implied by the Lipschitz constant. Let $\lambda_K = \lambda_{\min}(K)$ and $\lambda_\Lambda = \lambda_{\min}(\Lambda)$.

Proposition 1: Under A1–A4, the closed loop formed by the inner loop (12) and the outer loop (10) satisfies, for all $t \ge 0$:

(i) the inner-loop error is bounded by

$$\|s(t)\| \le e^{-\lambda_K t}\,\|s(0)\| + \frac{\bar{d}}{\lambda_K}\left(1 - e^{-\lambda_K t}\right), \tag{15}$$

so that (12) is input-to-state stable [49] with respect to $\tilde{d}$ with linear gain $1/\lambda_K$;

(ii) in swerve mode, the configuration error is bounded, for $\lambda_K \neq \lambda_\Lambda$, by

$$\|e(t)\| \le e^{-\lambda_\Lambda t}\,\|e(0)\| + \frac{e^{-\lambda_K t} - e^{-\lambda_\Lambda t}}{\lambda_\Lambda - \lambda_K}\,\|s(0)\| + \frac{\bar{d}}{\lambda_K \lambda_\Lambda}\left(1 - e^{-\lambda_\Lambda t}\right), \tag{16}$$

with the second term replaced by $t\,e^{-\lambda_\Lambda t}\|s(0)\|$ when $\lambda_K = \lambda_\Lambda$, and therefore $\limsup_{t\to\infty} \|e(t)\| \le \bar{d}/(\lambda_K \lambda_\Lambda)$;

(iii) in differential-drive mode, if the nonholonomic outer loop $\dot{e} = g(e,t) - \bar{S}(q)\,s$ is input-to-state stable with respect to s with asymptotic gain κ, then $\limsup_{t\to\infty} \|e(t)\| \le \kappa\,\bar{d}/\lambda_K$.

Proof: The inner-loop dynamics (12) do not depend on e, so we bound s first and then propagate the bound through (10).

For (i), let $V = \frac{1}{2}\|s\|^2$. Along (12), $\dot{V} = -s^\top K s + s^\top \tilde{d} \le -\lambda_K \|s\|^2 + \bar{d}\,\|s\|$, so that $\frac{d}{dt}\|s\| \le -\lambda_K \|s\| + \bar{d}$ for $\|s\| \neq 0$, and the comparison lemma [50, Lemma 3.4] gives (15). Replacing $\bar{d}$ by $\sup_{\sigma \le t} \|\tilde{d}(\sigma)\|$ gives the ISS estimate with gain $1/\lambda_K$. The nominal part of (12) is contracting with rate λ_K [51], as in [5]; since $\tilde{d}$ depends on the state history through z_t, we state the result as ISS rather than as contraction of the learned closed loop.

For (ii), the variation-of-constants formula for (10) gives

$$e(t) = e^{-\Lambda t} e(0) - \int_0^t e^{-\Lambda(t-\sigma)}\,S(q(\sigma))\,s(\sigma)\,d\sigma. \tag{17}$$

Since Λ is diagonal with positive entries, $\|e^{-\Lambda t}\| = e^{-\lambda_\Lambda t}$, and since $S(q)^\top S(q) = I$, $\|S(q)s\| = \|s\|$. Taking norms and substituting (15) under the integral, we obtain $\|e(t)\| \le e^{-\lambda_\Lambda t}\|e(0)\| + \int_0^t e^{-\lambda_\Lambda(t-\sigma)}\left(e^{-\lambda_K \sigma}\|s(0)\| + \bar{d}/\lambda_K\right) d\sigma$, and evaluating the two integrals gives (16). The first two terms vanish as $t \to \infty$, which gives the asymptotic bound. Since e and s are bounded, $v = v_r + s$ and $\dot{v}_r$ are bounded under A1, so that (11) is well defined along the closed-loop trajectory.

For (iii), the outer loop admits an ISS estimate $\|e(t)\| \le \beta(\|e(0)\|, t) + \kappa \sup_{\sigma \le t}\|s(\sigma)\|$ with β of class $\mathcal{KL}$, and s is ultimately bounded by $\bar{d}/\lambda_K$ from (i); the cascade is therefore ultimately bounded by the product of the bounds [50, Sec. 4.9]. ■

The ultimate bounds are proportional to $\bar{d}$, so that the learned $\hat{d}$ reduces the ultimate bound at fixed gains provided that a smaller uniform bound $\bar{d} < D$ is established for it. The gains have separate roles: λ_K sets how fast the realized twist follows the reference and how much of $\tilde{d}$ reaches the outer loop, and λ_Λ sets how fast the configuration error decays for a given inner-loop error. The risk-aware planner of [32] bounds the same residual-model error with conformal prediction; we have not established a mapping between $\bar{d}$ and that disturbance set, and we report the two separately.

Implementation

We identify the time constants and DC gains of (7) by a least-squares fit of the zero-order-hold discretization $v[t+1] = \alpha v[t] + \beta u[t]$ on rigid ground, with $\tau = -\Delta t/\ln\alpha$ and $k = \beta/(1-\alpha)$, from translation and point-turn step commands excited separately. For NOVA in swerve mode the identified values are $\tau_v = 1.007$ s for the two translational channels and $\tau_\omega = 0.820$ s for yaw, with $k_v = 0.566$ and $k_\omega = 0.290$, where the command u in the fit is the twist mapped through (4) to duty cycles. The inner loop and residual model are specified for 20 Hz operation. The current NOVA implementation runs only the kinematic outer loop at 20 Hz. The initial inner-loop gain is $K = \mathrm{diag}(1,1,1)\ \mathrm{s}^{-1}$, so that the nominal inner-loop error decays at the rate $\lambda_K = 1\ \mathrm{s}^{-1}$, and the outer-loop gain is $\Lambda = \mathrm{diag}(0.5, 0.5, 0.5)\ \mathrm{s}^{-1}$.

The current Pico/MDD10A interface accepts signed PWM duty commands rather than torque or wheel-speed commands; the commanded twist is mapped through (4) to steering positions and nominal wheel speeds, and the wheel speeds are converted to bounded duty cycles by a fixed scale. The controller that has run on NOVA is the kinematic outer loop closed on the motion-capture pose at 20 Hz, implemented in the nonholonomic form with gains $k_x = 1$, $k_y = 2$, $k_\theta = 1$, with its reference twist passed directly to the actuation layer; this is the controller of the circular-trajectory experiment of Section 5. The inner loop (11) with the learned residual has been closed on the Leo Rover (Section 5).

Remark 1 (Differential Drive Mode) Differential-drive mode is selected by a launch-time parameter of the actuation layer, which sets the four servo goal positions to the zero-steering calibration values and replaces (4) with (6); NOVA then has the kinematics of the skid-steer Leo Rover of Section 3. The mode is a fallback after a steering-servo fault only when the faulty servo can still hold its wheel at zero steering or the wheel can be pinned straight.

Remark 2: The Leo Rover is NOVA in differential-drive mode with $\delta_i \equiv 0$ enforced mechanically. Note that while the Leo Rover shares the two-input planar model structure used for NOVA in differential-drive mode, it still requires separately identified dynamics and a separately trained residual model. The cascade control structure applies to it, and it is the platform on which the inner loop and the residual model were developed and validated [31]. On the Leo, the twist is $\bar{v} = [v_x, \omega]^\top$, the inner model is the two-channel $\bar{A}_n$, $\bar{B}_n$ identified by the same zero-order-hold fit through the vendor velocity interface, the residual $\hat{d}(z_t)$ is the two-channel model trained on the data of Section 5 with the fourteen-channel proprioceptive vector of [31], the control law (11) is unchanged with $K \in \mathbb{R}^{2\times 2}$, and the outer loop is the nonholonomic controller, so that part (iii) of the proposition applies. With all wheels straight, the slip residual ε_i of (5) detects only longitudinal inconsistency between the wheels, and lateral slip during a turn must be observed from the IMU or from the motion-capture twist, as the fault estimator of [25] does with its exteroceptive anchor. Since the Leo and NOVA differ in wheels, mass, and actuation as well as in steering, the direct test of the effect of steering on the residual d is to run NOVA in its two modes under matched terrain and reference.

5. Experiments Conducted in the Facility

In this section, we describe the experiments that have been conducted in the facility to date, stating for each the platform on which it was run and the conditions under which the result holds. The facility supports three modes of use: individual

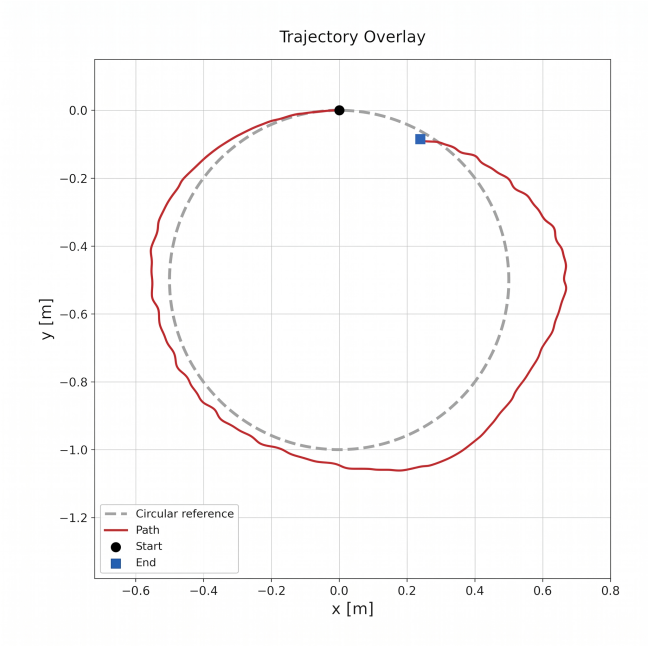
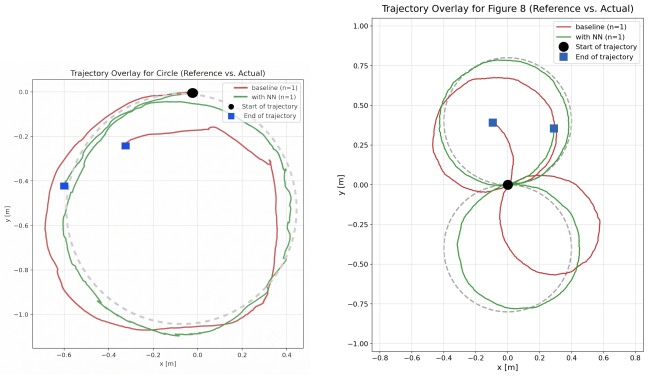


Figure 11: NOVA tracking a circular reference on the gravel bed under the baseline kinematic outer loop with motion-capture feedback.



(a) Circle. (b) Figure eight.

Figure 12: Closed-loop tracking on the Leo Rover: baseline contraction controller (red) and the same controller with the learned residual as feed-forward correction (green); reference in dashed grey.

subsystems such as visual-inertial odometry, terrain segmentation, or the fault estimator are tested under controlled lighting and terrain changes with the ground-truth state available for scoring; closed-loop navigation is evaluated with the complete stack, either under motion-capture state or under onboard estimation; and multimodal data are collected for learning-based methods with the metadata retained so that training and test conditions are separated by environment (a held-out slope, a held-out lighting azimuth, a held-out surface preparation) rather than only by random sample, which is the separation that a distribution-shift study requires.

NOVA: Tracking Control on Regolith

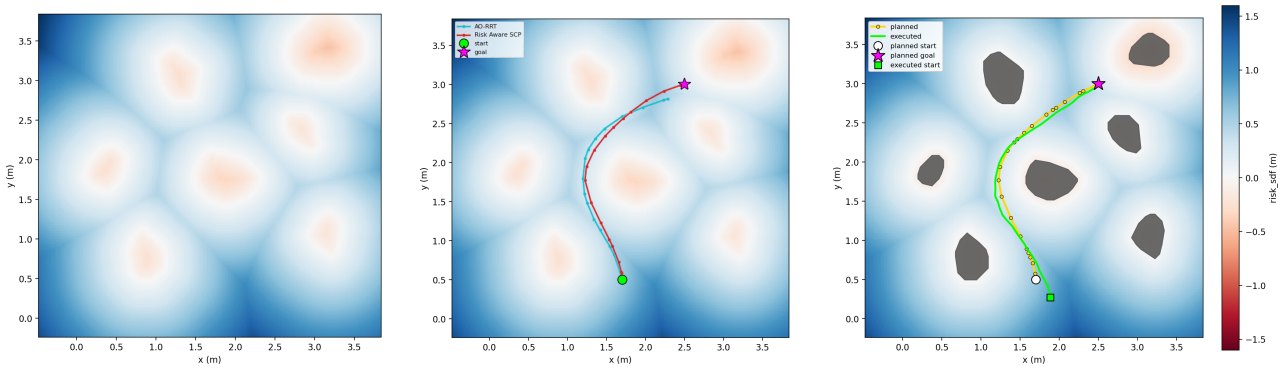
An initial circular trajectory-tracking experiment was conducted with NOVA on the gravel bed in swerve mode, with the baseline kinematic outer loop of Section 4 closed on the motion-capture pose at 20 Hz and the reference twist mapped through (4) to steering positions and open-loop drive duty cycles (Fig. 11). The rover followed the circular reference through its independently steered and driven wheel modules, which demonstrates the integrated operation of the sensing, actuation, and software of Section 3 on the gravel. The controller does not model the terrain and the slip, which leads to a large tracking error (position error of up to approximately 0.20 m with respect to the motion-capture reference). The baseline controller does not explicitly compensate for terrain-induced residual dynamics. The following subsection reports residual-learning and closed-loop tracking experiments on the Leo Rover. Transfer of the residual-compensated controller to NOVA remains future work.

LEO rover: Learning-Based Control on Regolith

The proprioceptive residual-learning controller of [31] was developed on the Leo Rover in the facility. The dataset consists of nine driven traversals recorded at 20 Hz: seven on the flat portion of the bed (straight runs, arcs, circles, and figure-eights covering the operating envelope, approximately 32,000 samples) and two crossing the sloped portion of the bed in different orientations. The flat data were split chronologically per traversal into training and validation sets, while the two slope traversals were held out entirely as an out-of-distribution test set, so that the network never saw slope data during training. The residual target $d = \dot{v} - A_n v - B_n u$ was computed from the 100 Hz Vicon twist, and the nominal parameters (k, τ) were identified from the flat-terrain training traversals. On the held-out flat data the learned residual reduced the root-mean-square residual from 0.169 to 0.053 m/s^2 in the longitudinal channel and from 0.406 to 0.125 rad/s^2 in the yaw channel, which is a 68.7% and 69.2% reduction relative to the nominal model alone; on the unseen slopes the reduction was 49.1% and 56.7% overall, and 59.9% and 65.1% on the lateral-slope traversal, where the gravity projection on the tilted bed is largest. In closed loop on the Leo Rover, the same contraction-based controller with the learned residual added as a feed-forward disturbance term tracked circle (Fig. 12a) and figure-eight (Fig. 12b) waypoint references on the gravel more tightly than the nominal controller, using only onboard sensing for pose feedback. The next step is to transfer the same controller to NOVA using the control law of Section 4.

Risk-Aware Planning and Control Under Uncertainty

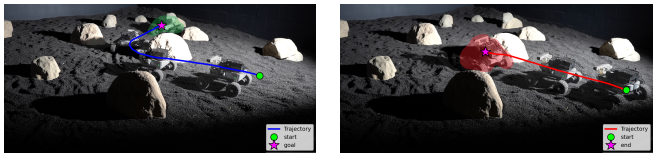
The architecture of [32] accounts for uncertainty that enters the navigation task through two channels: uncertainty that can be quantified before planning (mapped obstacle boundaries and learned residual dynamics) and uncertainty that is revealed online (sensor bias). Residual wheel–terrain dynamics are learned by a fully connected network from motion-capture data collected on the bed, and the error of the learned model is bounded with split conformal prediction [52] to build a disturbance set; obstacle uncertainty is modeled by inflating each mapped boundary and combined with slope and roughness scores into a risk map and a conditional value-at-risk (CVaR) [53] keep-out field. Planning proceeds in two steps: a sampling-based kinodynamic planner (AO-RRT [54]) generates a dynamically feasible, risk-aware trajectory whose edge cost combines control effort and an empirical CVaR over closed-loop rollouts under the disturbance set, and the resulting trajectory seeds a sequential convex programming refinement [55] that jointly minimizes control effort and risk. The closed loop is shown in Fig. 15. The trajectory is tracked by a model predictive controller solved by sequen-



(a) (b) (c)

Figure 13: Hardware experiment of the risk-aware planner [32] in the facility. (a) Risk-inflated signed distance field of the map. (b) AO-RRT seed and the risk-aware SCP trajectory refined from it. (c) Planned trajectory and the trajectory executed by the rover, with the nominal rock footprints in grey. The colorbar in (c) applies to all three panels. The rover executing this trajectory is shown in Fig. 14a.

tial quadratic programming, which propagates the estimator covariance along the horizon, tightens the keep-out constraint by the corresponding confidence margin, and adds a stopping-distance terminal control-barrier-function constraint [56] for recursive feasibility [57], so that infeasibility triggers a safe stop or a replan.



(a) Risk-aware control. (b) Risk-neutral control.

Figure 14: Leo Rover with a ZED 2i stereo camera and a Jetson AGX Orin on the gravel bed, with the Vicon system providing ground truth. Under the same injected sensing bias, the risk-neutral controller collided in the trial shown in (b), while the risk-aware controller in (a) completed the route.

In the hardware demonstration, the architecture was run on the Leo Rover fitted with a ZED 2i stereo camera (1920×1080 at 30 fps) and a Jetson AGX Orin on the gravel bed with an average 3% grade, with the Vicon system providing ground truth and with sensing bias emulated by adding a fixed offset between the motion-capture pose and the pose given to the controller. The nominal MPC was run in 9 trials and collided in 5 of them, violating the keep-out region in 2 more; the risk-aware MPC was run in 16 trials and did not collide or violate the keep-out region in any of them, with a minimum motion-capture clearance of 0.288 m, completing the route in 5 trials and stopping itself short of the goal in the remaining 11, in which the injected bias exceeded the available clearance. The nominal controller was run in 9 trials and the risk-aware controller in 16, so that the two conditions are compared as sets of trials rather than as matched pairs. Figure 13 shows one risk-aware run: the risk-inflated distance field built from the stereo map (a), the AO-RRT seed and the SCP trajectory refined from it (b), and the planned and executed trajectories over the rock footprints (c); Fig. 14 shows the rover on the bed during this run (a) and during a risk-neutral run under the same injected bias that ended in a collision (b). A confidence study at an injected bias of 0.10 m, with the position uncertainty reported to the controller set to a standard deviation of 0.20 m, showed that the controller completes the path at confidence levels of 0.80 and 0.85 but stops after 17% and 14% of the route at 0.90 and 0.95, which quantifies the conservatism that the confidence parameter trades against task completion. The facility will be used to repeat this study as the obstacle layout, the lighting, and the surface preparation are varied.

Fault Detection and Resilience

The invariant adaptive fault estimator of [25] models actuator-side and sensor-side faults as biases on the body twist, fuses wheel-encoder velocities and gyro yaw rate in an invariant extended Kalman filter [58] on $SE(2)$, and runs a fault-tolerant invariant adaptive observer whose invariant error between the estimate and a low-rate exteroceptive pose anchor drives an adaptation law for the fault bias, with observer and adaptation gains synthesized offline by a convex linear matrix inequality so that the composite error is uniformly ultimately bounded. A fault detection and identification logic then classifies the estimated bias into velocity, wheel, actuator, or exteroceptive faults, and the recovered bias is used to compensate the commanded twist. In simulation, 1000 Monte Carlo trials on a four-wheel mecanum model with one fault per trial gave observer detection rates of 66.5–100% and identification rates of 92.2–100% across the six fault channels, with the recovery reducing the position error from 108 m to 2 m for a longitudinal-velocity fault.

On hardware, the observer was run on the Leo Rover on the basalt simulant bed at 30 Hz, with the Vicon pose, subsampled to 4 Hz, substituting for the GPS and sun-sensor anchors that a planetary rover would carry. The rover traced four circles with a fault injected at a random time during the second circle, with ten trials per fault type (velocity, yaw rate, encoder, actuator, and exteroceptive) for 50 trials in total. Velocity, yaw rate, and encoder faults are injected in the encoder stream and actuator faults are injected in the rover's actuation stream. The exteroceptive fault is injected by corrupting the exteroceptive anchor before it is utilized in the fault-tolerant observer. Across the 50 trials the observer detected the injected fault in 100% of trials and identified its type in 94%, with a median detection latency of 0.8 s, and the recovered fault magnitude reported in [25] was 104.1% and 99.2% of the injected magnitude for the additive velocity and yaw-rate faults, while a chi-squared residual test on the invariant EKF detected 72% of the faults but fired before the injection in 92% of those cases. These results are for

faults injected into the encoder, actuation, or anchor streams while the rover drives on the gravel; the detection of naturally occurring slip, which the facility can provoke with loose patches, slopes, and obstacle contact and score against the motion-capture twist, has not yet been evaluated. In the proposed closed loop, a detected fault would update the planner's uncertainty model or trigger a mode switch.

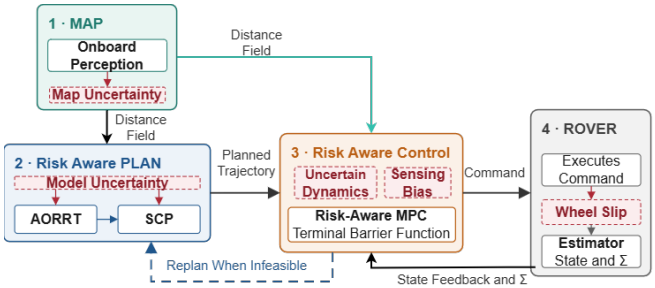


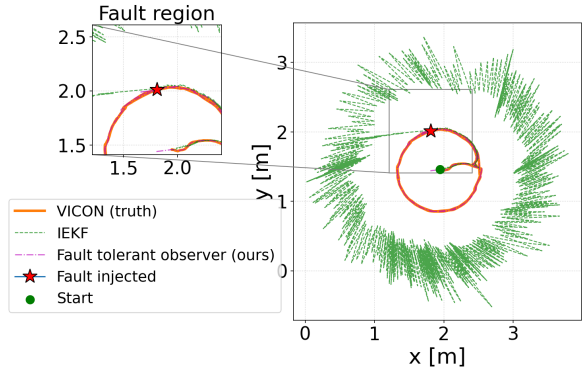
Figure 15: Closed-loop architecture of the risk-aware planning and control pipeline of [32] as run on the Leo Rover.

Figure 16 shows a representative trial with an additive fault injected into the longitudinal body velocity v_x^B of the encoder stream. The fault-tolerant observer follows the Vicon ground truth, whereas the invariant EKF, which carries no fault state and fuses the corrupted encoder velocity, diverges after the injection on the second circle (Fig. 16a). In this trial the observer detected the fault 0.53 s after it occurred, the fault estimate $\hat{d}$ converged to the injected value d and remained within the band shown in Fig. 16b for the rest of the run, and the χ^2 baseline raised a false detection 5.55 s before the injection.

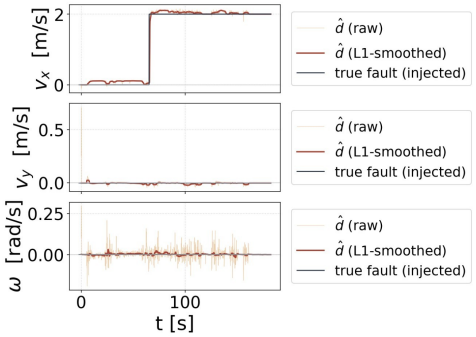
Figure 17 shows a representative trial with a multiplicative fault injected into the exteroceptive anchor. The fault-tolerant observer follows the Vicon ground truth while the invariant EKF, which fuses the corrupted anchor, diverges after the injection; the fault is detected and identified as exteroceptive, but its magnitude is not recovered, since the adaptation law estimates biases on the body twist and not on the anchor.

Multi-Modal Mobility

Three multi-modal mobility studies have used the facility bed or the lab's single-wheel testbed; the first two were run on the facility bed and the third was not. First, the terrain-adaptive humanoid locomotion policy of [44] computes foot–terrain contact with a three-dimensional resistive-force-theory model [59], [60] that treats each foot as a grid of collider points and sums the resistive stress on the submerged points into a net wrench at the ankle, implemented in NVIDIA Warp so that 4096 parallel environments train in Isaac Lab [61] on a single GPU. The policy is trained by proximal policy optimization [62] in a teacher–student setup in which a variational terrain encoder reconstructs a normalized media stiffness from proprioceptive history, and the student is distilled with a temporal convolutional encoder. In simulation against a material-point-method reference, the 3D-RFT-trained policy achieved an 85% success rate on deep, low-friction sand, where heuristic granular models reached at most 8%, and 100% success across basalt, sand, and poppy-seed terrains, where the rigid-contact policy failed on sand and poppy seed. On hardware, the Unitree G1 walked and ran on the basalt bed and on dry sand at commanded velocities of 0.5 to 2.5 m/s, while a rigid-contact baseline failed to maintain balance on basalt at all commanded velocities. The



(a) Trajectory during the four-circle maneuver, showing IEKF divergence after injection.



(b) Fault estimate $\hat{d}$ versus the true injected fault d.

Figure 16: Hardware response of the Leo Rover to a v_x^B fault.

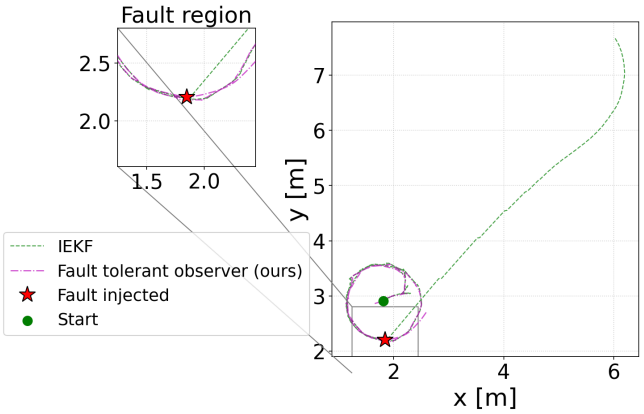


Figure 17: Trajectory during the four-circle maneuver under a multiplicative exteroceptive-anchor fault, showing the divergence of the invariant EKF and the fault-tolerant observer following the ground truth.

policy raised its swing-foot clearance from a nominal 9 cm to 16 cm as the decoded stiffness dropped on entering softer material.

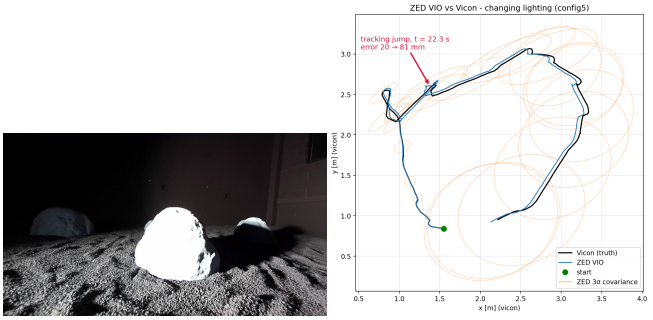
Second, the DUALITY wheel-leg rover [26] was tested over several weeks on the simulant bed on slopes built from 10° to 30°: in each test the rover was first driven up the slope in its rolling mode, and the crawling gait was then engaged to evaluate traction and slippage. The rolling gait became unreliable beyond 20°, usually partway up the slope, at which point the crawling gait was engaged and completed the

traverse, and in the most recent session the rover climbed a 30° slope using a turtle-style gait in which all hips are moved back, the chassis is lowered onto the knees, the wheels are rolled backward while the hips move forward, and the chassis is raised. The wheel and grouser sizing for the prototype was derived from Bekker-type drawbar-pull relations [63] with Apollo-era lunar soil parameters, and the facility tests were the primary validation because, as the report notes, analytical regolith-interaction models are inconsistent for small prototypes.

Third, the GROVER variable-height grouser wheel [45] was evaluated on a single-wheel gantry testbed in the lab over four surfaces of increasing granularity (vinyl, sand, pea gravel, and coarse red rock) at six grouser heights from 0 to 17.5 mm, with 25 trials per configuration for 600 trials in total. The optimal grouser height varied with the terrain (3.5 mm on vinyl, 7.0 mm on gravel and coarse rock, and 17.5 mm on sand), and the optimal height reduced slip by 30–58% relative to the baseline height, with a 77.4% reduction in both travel time and energy on sand, which follows a power-law relation between optimal height and granule diameter. These results were obtained on the gantry testbed with its own surfaces, not on the facility bed, and they motivate the full-vehicle grouser-adaptation tests on the bed that are planned once the wheel is integrated onto a rover. The humanoid and wheel-leg experiments demonstrate two locomotion modes on the basalt bed.

Visual-Inertial Odometry Fault Under Directional Lighting

In the facility, with a single lamp at low elevation and the remaining lamps off, the bed presents the stereo camera with long hard-edged shadows, lit rock faces that saturate at the exposure needed for the shadowed ground, and a dark surround (Fig. 18a). In a preliminary test of the ZED 2i visual-inertial odometry on the Leo Rover under this lighting, we observed discrete jumps in the pose estimate relative to the Vicon ground truth, one of which is marked in Fig. 18b, and a growth of the reported state covariance over the run until the estimate was no longer usable for control. Shadow boundaries that move with the rover and the loss of texture in saturated and shadowed regions are possible causes, but the test did not include the diagnostics (feature tracks, exposure, or inlier counts) that would attribute the jumps to either, so that we report the behavior as an observation.



(a) View from the ZED 2i camera.(b) Visual-inertial state estimate compared with the Vicon ground truth; the arrow marks a jump in the estimate.

Figure 18: (a)Single-lamp low-elevation lighting produces long, sharp shadows and saturated lit faces in the camera view ; (b) under this lighting the visual-inertial estimate showed a discrete jump relative to the Vicon ground truth.

6. Scope, Limitations, and Planned Characterization

The facility is an intermediate robotics test environment and has limitations. The facility supports repeatable physical experiments with reconfigurable granular terrain, adjustable directional illumination, and an independent motion-capture pose reference. Since the same terrain geometry can be observed under different source directions, the autonomy stack can be tested on differently prepared surfaces and on different mobility platforms, and software changes can be compared without relying on naturally varying outdoor illumination, all with the true state of every robot known.

Limitations.—Tests are conducted under Earth gravity and atmospheric conditions; the 3–5 mm basalt gravel does not reproduce the particle-size distribution or cohesion of lunar regolith and has not been fully geotechnically characterized; the lighting system provides adjustable directional illumination but does not reproduce the Sun's spectrum, angular size, or irradiance, and the illuminance it produces at the bed has not been measured; the $5.8\ \mathrm{m} \times 4.0\ \mathrm{m}$ area limits long-distance navigation and the 0.30 m fill depth bounds the relief that can be shaped in the bed; and the motion-capture system introduces retroreflective markers that must be kept out of the onboard camera's field of view or masked in the perception pipeline.

The next stage of facility characterization will focus on: measured two-dimensional illuminance maps for representative source configurations, reported as E_{mean}, E_{min}, E_{max}, and the coefficient of variation $\sigma_E/E_{\mathrm{mean}}$, together with their run-to-run repeatability after resetting lamp position, aiming, and dimming; the bulk density and angle of repose of the installed simulant; motion-capture accuracy across the bed under the directional lighting and the time-synchronization error between the Vicon stream and the onboard logs; baseline NOVA localization and navigation performance under nominal conditions, with and without motion-capture state in the loop; and controlled comparisons as lighting, terrain, platform, and autonomy components are varied one at a time.

7. Conclusion

In this paper, we discussed the development of the Lunar Terrain Facility at the Georgia Tech AstroX for developing and testing learning-based autonomy on rovers, multi-modal robots, and collaborative multi-agent robotic systems. The facility consists of a $5.8\ \mathrm{m} \times 4.0\ \mathrm{m} \times 0.30\ \mathrm{m}$ bed of 3–5 mm Merriam Crater basalt gravel with reconfigurable rock and relief features, four lamps with Fresnel optics for adjustable directional illumination, and a 16-camera motion-capture system that supplies a pose reference for every robot and marker-equipped object in the bed at 100 Hz. We described the mechanical, electrical, and sensing subsystems of the NOVA rover, the other platforms exercised on the bed, and the ROS 2 software architecture that allows the estimation, planning, and control modules to be exchanged, and we presented a kinematic model of the independently steered rover with a first-order actuation model and a learned terrain residual, together with a tracking control law that we showed to be input-to-state stable with respect to the residual-model error under the stated assumptions, with ultimate bounds proportional to the uniform bound on that error.

The experiments conducted in the facility to date establish the following. NOVA tracks a circular reference on the gravel under motion-capture feedback with its baseline kinematic loop. On the Leo Rover, a learned proprioceptive residual reduced the RMS residual-model error by 49–69% on held-out data; the risk-aware architecture completed five of sixteen trials and stopped safely in the other eleven, with no collisions, under injected sensing bias, where the nominal controller collided in five of nine; and the invariant adaptive observer detected all and identified 94% of fifty injected faults. The Unitree G1 walked and ran on the gravel, and the DUALITY rover switched from rolling to crawling on 10–30° slopes. A stereo visual-inertial estimator exhibited pose jumps and covariance growth under single-lamp low-angle lighting. The studies evaluated individual platforms, with the terrain, lighting, and pose reference recorded for all but the visual-inertial test; coordinated multi-robot operation remains future work.

Acknowledgements

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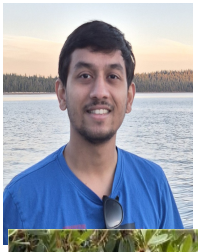

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